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Originality Detection Software in a Graduate Policy Course: A Mixed-Methods Evaluation of Plagiarism

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The authors used a mixed-methods approach to evaluate the use of Turnitin originality detection software in a graduate social work course. Qualitative analysis of student responses revealed positive and negative spent completing assignments, and the tone of the class. Quantitative analysis of students' originality scores indicated a short-term reduction in student plagiarism, although the full level of decline was not maintained in the long term. Implications for originality issues in social work education and social work practice are presented.

KEYWORDS plagiarism, social work, originality, social work education

Social workers use technology to facilitate professional education and to retrieve evidence of practice effectiveness (Swiefach & Heft-LaPorte, 2006). In 2005, the National Association of Social Workers collaborated with the Association of Social Work Boards to develop practice standards for technology for social workers who practice in the United States and Canada. The technology standards are an application of common values, ethics, and practice standards that were adapted for technology-driven environments. The ultimate goal is to serve as a guide to social workers who use technology

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and to encourage the promotion of technology competencies across practice areas including research using uniform skills and knowledge (National Association of Social Workers & Association of Social Work Boards, 2005). The standards include specific mention of originality as it relates to verifying the accuracy of online sources and thus establishing the credibility and limitations of the information (National Association of Social Workers & Association of Social Work Boards, 2005). To date, the technology standards do not address originality issues such as presenting online sources as original work in written assignments, grant applications, research, or professional presentations. This article seeks to identify some of the concerns related to originality of student work in an effort of the authors to shape this discussion.

Educators have identified issues associated with plagiarism and plagiarism detection in social work education. Central among these issues are questions about whether students understand citation guidelines and consequences of plagiarism (Vernon, Bigna, & Smith, 2001). Furthermore, there is often a lack of consensus about what constitutes plagiarism (Collins & Amodeo, 2005). Educators in other fields also identified student citation errors as a major contributing factor behind acts of plagiarism in higher education (Kendall, 2005). Others indicate that issues that contribute to plagiarism include paper-for-sale sites on the World Wide Web and basic word processing packages that allow people to cut-and-paste text from online sources to use in their projects (Lyon, Barrett, & Malcolm, 2004; Savage, 2004).

Social work educators have put forward methods of instruction and organizational policies that can enhance student compliance with academic honesty standards. These models can be adapted to accommodate technology-related dishonesty and the use of technology in detecting plagiarism. For example, Saunders (1993) advised social work programs to clearly inform students about what actions constitute plagiarism and the consequences for those actions. Saunders also argued that students will benefit if they understand the disciplinary process and the findings of prior cases. Collins and Amodeo (2005) expanded on these policy recommendations by proposing that faculty meet with the students to review the submitted work alongside the text from which the plagiarized passages may have originated. They also suggested that social work programs would benefit by establishing levels of severity of student misconduct that are related to the students' understanding of their actions and the extent of plagiarism in the assignment.

Gibelman, Gelman, and Fast (1999) referred to the process of submitting products purchased from online sites as "cybercheating" (p. 1). They advised social work faculty to modify their assignments substantially in a way that prevents students from drawing on other peoples' work. Vernon and his colleagues (2001) discussed cybercheating and ways that social work faculty could identify it in student writing. They suggested that students who

incorporate unoriginal material in their papers often have inconsistency in writing where one section may be unusually clear and well researched. They also suggested that papers that include cybercheating may be well researched with materials incorporated from sources not available at the university or that the well-researched paper maybe off topic.

In response to growing concerns about plagiarism, universities are using originality detection software, such as Turnitin, the technology used for the course in this study. Turnitin is the brand name of a plagiarism detection system that identifies portions of submitted papers that match sources from the Internet, including websites, journals with online access, and previously submitted student papers. Faculty submit the student papers to the Turnitin system and receive a report that indicates the portion of the paper that matches known sources by highlighting the matched sections and listing the corresponding source (Sheridan, Alany, & Brake, 2005).

Several evaluations of the Turnitin system indicated limited success in decreasing incidents of plagiarism. Pharmacy students in New Zealand (Sheridan et al., 2005) reported that using the software improved their citation rate and that they were more likely to create original works when using Turnitin. Similarly, faculty and students who responded to a university-wide evaluation of the system in Australia (Savage, 2004) and in New Zealand (Gulik & Tippin, 2004) reported that they believed using plagiarism detection software was a deterrent to cheating among students. This evaluation focused on MSW student response to Turnitin originality software with specific attention to whether it decreased incidents of plagiarism and how the students perceived their experience.

COURSE DESCRIPTION

The evaluation class was an advanced policy analysis course for students specializing in health and mental health care. This was a master's-level class in which students applied federal, state, and local policies to practice settings in health and mental health environments. Health topics included social justice, financing, stakeholder political agendas, care rationing, human resource management, and supervision. Specific mental health critiques included involuntary care, community care, and mental health reform. The class also examined ethical and legal issues affected by standards of practice and practice style. Included in the legal examination were consent, confidentiality, reporting harm, impairment, fraud, and regulations regarding licensure and professional competency.

The instruction methods included a variety of face-to-face techniques including discussion, debate, student presentations, and brief lectures. These activities included a formal discussion in the beginning of the semester where students debated whether health care was a human right or a

consumer choice. Students ended the semester with a mock licensing board hearing in which they acted out roles in a case that demonstrated common social work licensing issues. In addition, students met in groups during weekly class meetings to examine readings, resolve case examples, and share their solutions with the class for further examination and a broader comparison of policy resolutions.

The course included three individual written assignments that required students to connect policy to practice. The first assignment included a combined presentation and paper. The students wrote and presented a summary of a paper that examined the weekly topic from the perspective of significant stakeholders: consumers, hospitals, pharmaceutical companies, insurance carriers, employers, or social workers. The second paper examined social work supervision from a management perspective. This paper was designed to help students understand the interconnection between supervision and management in field settings. The final paper examined health care fraud, consequences to agencies and providers, and management strategies designed to ensure compliance with state and federal policies.

The students were expected to submit a hard copy and an electronic copy of their assignments. The professor submitted their papers to Turnitin for an originality evaluation. Papers that were not submitted in both formats received unsatisfactory scores. Students were also notified in the course outline that any papers that included improperly referenced material, or that consisted of more than 20% quoted materials, would be graded as unsatisfactory. Students who had questions regarding written assignments were encouraged to contact the instructor during office hours or via e-mail for consultation on their papers before the assignments were due. The university institutional review office approved this study before the evaluation semester with the understanding that the course instructor would not be involved in reviewing the data until the class was completed and grades were final.

METHODOLOGY AND RESULTS

Students

This study took place in a CSWE-accredited combined BSW/MSW social work program. Twenty-two graduate students were enrolled in this course and participated in the Turnitin evaluation. The sample included 18 Caucasians (3 men and 15 women), 3 African Americans (all female), and 1 Middle Eastern male. These students included 16 advanced standing students (72.7%). These students were eligible to complete an MSW in 11 months because they had earned a BSW from a CSWE-accredited university. Of students, 8 (36.3%) identified their practice focus as health care, 7 (31.8%) as addiction, 5 (22.7%) as mental illness, and 2 (9.1%) did not identify a specialty emphasis in their current field placement or for their future practice.

Research Questions

To evaluate these students' reactions to the Turnitin software, a mixed-method study was conducted. The methodology was guided by two research questions.

Question 1: Does the repeated use of Turnitin reduce student inclusion of plagiarized content in course papers?
Question 2: What positive and negative effects do students report from the use of Turnitin?

The first research question focused on student use of nonoriginal material. We assumed that the use of the software may promote originality in student writing through two mechanisms. First, Turnitin may serve as a deterrent to the intentional use of nonoriginal materials in papers, because it identifies plagiarism for the professor. Second, Turnitin may facilitate student understanding of plagiarism, therefore improve their writing. We also assumed that students who are unintentionally committing plagiarism will become aware of their mistakes after the first Turnitin experience. The score was examined by students with the instructor who also tutored students on alternative writing strategies.

The second research question focused on student perceptions. Of particular interest were the students' perceptions of Turnitin's effect on their experiences in the course. We assumed that if students improved their understanding of plagiarism, then they might find the experience beneficial. Also, we suggested that students might perceive the software as a tool that discourages dishonesty, therefore assuring that grades are balanced and reflective of student mastery. We believed that the answers to these two questions would create a modest yet comprehensive review of the Turnitin software in a graduate social work course.

Research Question 1

Method. To assess the effect of the Turnitin software on students' use of nonoriginal material in their course papers, a quantitative analysis of longitudinal data was conducted. The use of nonoriginal material was represented by the score that the Turnitin software produces and reports to the professor. For example, a Turnitin score of 4% would indicate that 4% of the paper is composed of nonoriginal material. This variable (represented by the Turnitin score) was tracked across three assigned papers in a Fall 2006 policy course. The longitudinal design, with three time points, allowed us to assess for changes across time in a student's use of nonoriginal material. The scores from 22 students were analyzed using paired sample *t* tests. Comparisons were made between the students' first and second paper scores to assess for the presence of a short-term effect. Comparisons were made between

the students' first and third paper scores to assess for the presence of a long-term effect. Results are reported in the form of class means and *t* test computations.

Results. The mean Turnitin score decreased in the short term and the long term. The average student received a Turnitin score of 11.05 on the first paper, indicating that 11.05% of the 4–5-page paper was composed of nonoriginal material. In general, a score of 11 would signify that a paper had several sections of multiple sentences that were quoted without citation. On the second paper, the average Turnitin score fell to 6.62%. The use of nonoriginal material in the third and final paper was lower, on average, than that found in the first paper. However, the average score on the final paper (8.76%) was higher than that found on the second paper. These differences, from the short-term (between Papers 1 and 2) and long-term view (between Papers 1 and 3), were not statistically significant. These results can be found in Tables 1 and 2. (Results from the paired-samples *t* tests and Wilcoxon signed ranks tests are reported.) The Wilcoxon tests were conducted because of violations of normality. While paired-sample *t* tests are robust to moderate violations of normality, the analysis was conducted in parametric and nonparametric forms to assess for any difference in results. Nonsignificant results were found when using both types of statistical tests.

Research Question 2

Method. Students completed a questionnaire at the end of the semester as part of the evaluation project. This instrument included a question about their perceptions of the strengths and limitations of the software. In particular, they were asked to "list all negative and positive effects of turnitin.com on your experience." We coded responses to this question and grouped them into categories that reflected similar themes within the positive and negative categories. The positive themes that emerged were that it reduces plagiarism and it improves writing. The negative themes included concerns that the software contributed to lower grades, spending more time than

TABLE 1 Short-Term Comparison of Paper Scores ($n = 22$)

Paper 1 mean Turnitin score	Paper 2 mean Turnitin score	Paired samples *t* test *p* value	Wilcoxon signed ranks test *p* value
11.05	6.36	.063	.118

TABLE 2 Long-Term Comparison of Paper Scores ($n = 21$)

Paper 1 mean Turnitin score	Paper 3 mean Turnitin score	Paired samples *t* test *p* value	Wilcoxon signed ranks test *p* value
10.90	8.76	.507	.945

expected writing papers, the suggestion of mistrust or anger, and issues with the technology, such as limits on student ability to submit papers independently, and possible software errors.

Data reduction was designed to produce information that could summarize student perception and experiences and to draw inferences from the evaluation of the software. Data reduction included quantifying student responses to generate descriptive statistics regarding the number and the percentage of responses in each category. The bin analysis model forwarded by Miles and Huberman (1994) was used to develop a conceptual framework for understanding student perception of the software for students with Turnitin scores that were above and below the class average score (Figures 1 and 2). This analysis method was applied to analyze data for conceptual differences across comparative groups of advanced standing and 2-year students, specialty (mental health, medical, substance abuse) and demographic characteristic of age, race, and gender. The comparison between students with low and high Turnitin scores was the only group comparison that revealed differences in perception of the software.

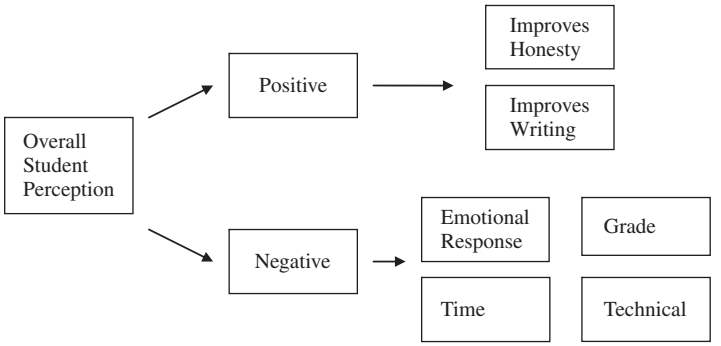


FIGURE 1 Conceptual Framework for Understanding Student Perception of Originality Software.

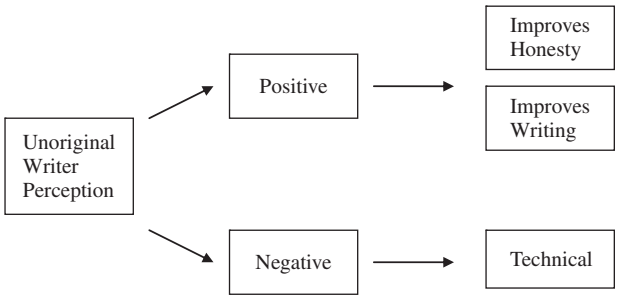


FIGURE 2 Conceptual Framework for Understanding Perception of Unoriginal Writing Scores of Originality Software.

Results. The positive responses indicated by the students included beliefs that Turnitin reduced plagiarism (29% of open-ended responses) and improved writing (21% of open-ended responses). Negative responses included concerns about the reporting process and access to the Turnitin software (18% of open-ended responses), emotional responses of anger directed at the instructor or suggestions that she mistrusted students (12% of open-ended responses), more time requirements to prepare papers to assure originality of writing (6%), and a possible negative effect on the student's grade (3%). Quotes that illustrate each of these themes can be found in Table 3.

A comparison of response patterns for students with high and low Turnitin scores was conducted to create a framework for student perception of Turnitin. Figures 1 and 2 convey the conceptualizations that represent the patterns of student perceptions between these two groups. As shown in Figure 1, students with low use of nonoriginal materials were the students

TABLE 3 Qualitative Themes by Writing Originality Group

Theme	Original writers	Unoriginal writers
Positive		
Improves honesty	"keeps people honest."	"keeps students honest."
	"eliminated the copying or reproduction of others' work without proper credit."	"keeps people from using other supports or thoughts without a citation."
Improves writing	"increased focus on correctly working my paper's content. (It) increased confidence in my ability to use my own thoughts. I feel like a better writer. I think I executed papers better and understood research better by writing papers, mostly from my perspective."	"after discussing why it was considered plagiarism, I became a much better writer and changed my style."
Negative		
Grades	"can get a bad grade on paper due to too much research."	
Technology issues	"I don't like that we don't always know if an instructor uses it or exactly what the results are. I don't like that our papers are apparently kept in Turnitin.com's database."	"was concerned that I plagiarized even though I gave complete credit to original source."
Emotional	"Sometimes it has made me hesitate to use references."	
	"I also feel it was a slight distrust on the part of the professor."	
Time	"I can't copy-n-paste my work at the last minute. I have to be on top of my APA formatting."	

Note. APA = American Psychological Association.

who expressed a broader range of themes in their negative perception of the software. They expressed concerns that Turnitin scores could influence their course grades and that concerns about the software caused them to spend more time writing the papers. The students with low Turnitin scores also expressed concerns that the use of the software suggested that the professor mistrusted them. They also shared concerns with students with high scores related to possible errors with the software. As shown in Figure 2, both groups expressed concerns about errors in the software that may flag their work as unoriginal.

LIMITATIONS

From our literature review, we believe that this study was the first to evaluate the use of the Turnitin software in a U.S. master's-level social work course. Because this research focused on a small purposive sample, the results are limited in their generalizability. While the study contributes to new understanding of the software's effects, replication is needed. It may also be beneficial to rotate topics in a second study to control for the difficulty level of subject matter and assignments. Another possibility would be to add an additional paper (Paper 4) to increase the sample size for the quantitative analysis of Turnitin scores.

The design also posed some limitations in the interpretation of the results. The longitudinal nature of the quantitative design was a strength. However, the design lacked a clear delineation of pre- and postbehavior. Students began each assignment knowing that their papers would be submitted to the Turnitin software. This knowledge may have led some students to reduce their use of nonoriginal material before the first data point (the first Turnitin score). Therefore, announcing the planned use of the software may have more effect than the results of this study were able to reveal. One difficulty in the examination of the effect of technology, such as Turnitin, is that the software functions as an assessment tool and a possible intervention. In essence, knowledge of the upcoming assessment (the Turnitin score) may be producing a testing effect by causing students to change their behavior before implementation of the project. To eliminate these threats to the design, future researchers would have to include a control group and use the assessment process without the control groups' knowledge. Although this type of design might aid in controlling the testing effect, its implementation would involve ethically problematic practices of deception that are not acceptable in social work research.

The qualitative results face the possible limitation of social desirability bias. Students often feel compelled to report attitudes that will be pleasing to their professors. To reduce the possibility of this bias, the qualitative survey was administered by a graduate assistant from another academic unit.

The students also were informed that their instructor would not be able to identify their personal responses and that all data would only be accessed after course grades were finalized at the end of the semester. Although this arrangement may have served to reduce social desirability bias, the threat cannot be completely eliminated.

DISCUSSION AND IMPLICATIONS

As this evaluation has shown, the percentage of nonoriginal material in these social work students' papers was fairly low. This is a promising finding. However small, there was still some unoriginal material detected. The reason behind the misconduct is unclear; it may be caused by laziness, disregard for ethics, or simply a lack of knowledge about plagiarism. What is known is that Turnitin plagiarism detection software does decrease student use of nonoriginal material, least in the short term. Unfortunately, through the application of qualitative analysis it has been shown that the use of such software can influence the perceptions of the students. These perceptions may lead to a more hostile work environment for the professor and a feeling of alienation among honest students.

The evaluation of the Turnitin software as a method for reducing the use of nonoriginal material yielded mixed results. The average student appeared to show compliance and change in the short term by reducing the percentage of nonoriginal material. Between the first and second papers, the Turnitin score reduced by nearly half. However, this change was not statistically significant. By the end of the semester, the use of nonoriginal material began to rise but still remained below the original level. These results cannot support the idea that the use of the Turnitin software will produce substantial long-term change in student use of nonoriginal material.

These results have many implications for social work educators. For example, Turnitin may decrease plagiarism but possibly at the cost of weakening the relationship between professor and student. The best students, that is students with the lowest Turnitin scores, seemed to be the most negatively affected by Turnitin. However, originality software may also help all students learn about ethical standards regarding dishonesty. To do so, it may be beneficial to have an introduction to plagiarism at the start of the semester where terms are clearly defined and the ethical issues are discussed specifically within the context of the social work profession. At the least, these findings suggest that graduate social work students may benefit from writing support being incorporated into their courses. Although this may take time out of class, in the long run, it may be worthwhile if the amount of plagiarism is reduced and professional behavior is favorably altered.

Social work is a field founded on ethics, so it is important to teach about all forms of ethical misconduct. Students, after all, need to learn ethical

expectations before entering the professional workforce. It is also relevant for professional social work standards to incorporate originality into technology standards to help guide students and practitioners. These standards should address cybercheating and the use of unoriginal work to ensure that the work they submit for grants, reports, research, and other professional-related activities is original material, not published or authored, in whole or part, by someone else. In particular, social workers who conduct research should ensure that authorship is accurately reflected and other standards for originality are understood and uniformly applied.

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